

Calculus – Group Discussion

Week 4 – Indefinite Integration Techniques

ADSAI & CS / PR12

Learning goals (Week 4)

- Compute basic antiderivatives using standard formulas.
- Understand the relationship between differentiation and antiderivatives.
- Apply substitution to evaluate integrals.
- Use integration by parts for products of functions.

Antiderivatives and indefinite integrals

An **antiderivative** of a function f on an interval I is a function F satisfying

$$F'(x) = f(x) \quad \text{for all } x \in I.$$

The **indefinite integral** of $f(x)$ is the family of all antiderivatives:

$$\int f(x) dx = F(x) + C,$$

where $C \in \mathbb{R}$ is the **constant of integration**.

Why does the constant appear? If $F'(x) = f(x)$, then also

$$\frac{d}{dx}(F(x) + C) = f(x)$$

for every constant C .

Basic antiderivative rules.

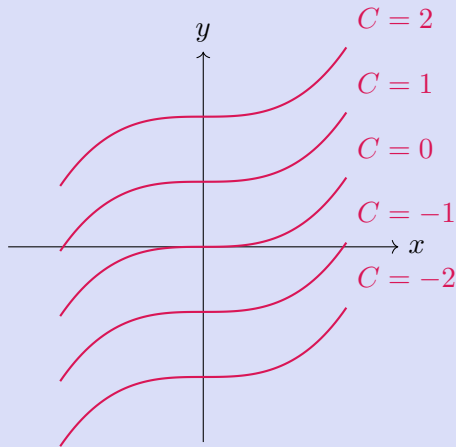
$$\int x^n dx = \frac{x^{n+1}}{n+1} + C \quad (n \neq -1)$$

$$\int \frac{1}{x} dx = \ln |x| + C$$

$$\int e^x dx = e^x + C$$

$$\int \sin x dx = -\cos x + C$$

$$\int \cos x dx = \sin x + C$$



$F(x) + C$ is the family of all antiderivatives with the same slope, but different vertical shifts for each $C \in \mathbb{R}$.

Integration by substitution

u -substitution $\iff$ Integral version of the chain rule.

$$\frac{d}{dx} f(g(x)) = f'(g(x))g'(x)$$

integrating both sides implies

$$f(g(x)) + C = \int f'(g(x))g'(x)dx.$$

Let $u = g(x)$, then $\frac{du}{dx} = g'(x)$ or in **differential form** $du = g'(x)dx$.

$$f(g(x)) + C = f(u) + C = \int f'(u)du.$$

Integration by parts

IbP $\iff$ Integral version of the product rule.

$$\frac{d}{dx}(u(x)v(x)) = u(x)\frac{d}{dx}v(x) + v(x)\frac{d}{dx}u(x)$$

integrating both sides implies

$$u(x)v(x) = \int u(x)\frac{d}{dx}v(x)\cancel{dx} + \int v(x)\frac{d}{dx}u(x)\cancel{dx}$$

$$u(x)v(x) = \int u(x)dv(x) + \int v(x)du(x)$$

This gives the **integration by parts** formula:

$$\int u(x)dv(x) = u(x)v(x) - \int v(x)du(x)$$

Key takeaways

- An indefinite integral is the family of all antiderivatives, so always include $+C$.
- Basic antiderivatives come from reversing standard derivative rules.
- Substitution is useful when the integrand contains an inside function together with its derivative.
- Integration by parts comes from the product rule and is helpful for products of unlike functions.
 - *Useful heuristic*: differentiate algebraic or logarithmic factors and integrate exponential or trigonometric factors
 - Sometimes you have to apply IbP more than once
 - Trade one integral for a hopefully easier one
- A good first step is always pattern recognition: basic rule, substitution, or by parts?